

Infrastructure Design for Vantage IT Solutions

Capstone Project

Neil Montances • Evan Warkentin
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One secure system must serve both offices

Vantage IT Solutions is a small startup that delivers network deployment, administration, support, and IT consulting. The infrastructure must support daily work while remaining practical for a small IT team to operate.

2

locations
HQ + branch

<30

employees
across both sites

1

Centralized
at headquarters

Core requirement: branch staff must reach approved HQ services without exposing management or user networks.

A focused device set supports the complete design

The hardware follows a clear role hierarchy:

- Cisco routers provide WAN edge and VPN termination.
- FortiSwitch 124F is the HQ Layer 3 core and VLAN gateway.
- Three Catalyst 2960 switches provide access switching.
- Linksys and TP-Link units operate as an access points.
- HP Probook 32 GB of RAM/1 TB as Proxmox host runs centralized services.

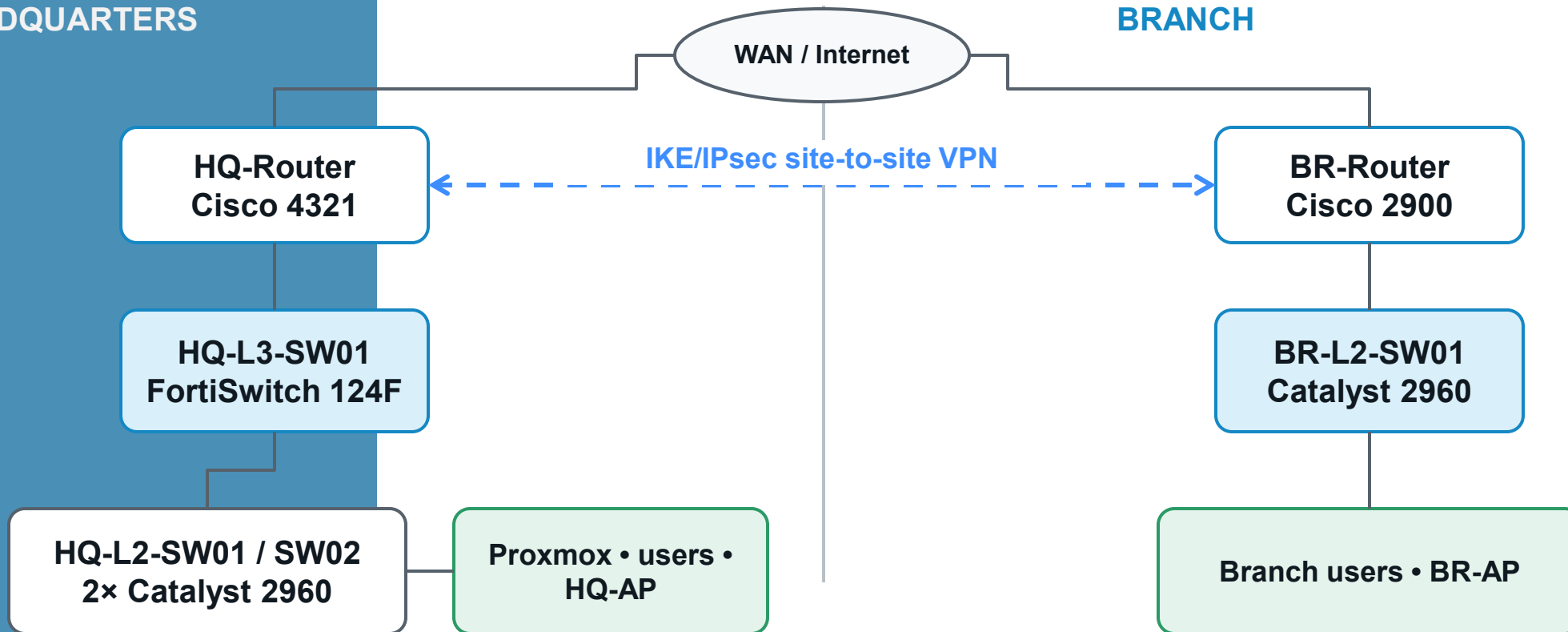
Asset Tag	Name	Image	Model	Category	Location
VIT-NET-001	HQ-Router		Cisco 4321 Integrated Services Router	Router	HQ
VIT-NET-003	HQ-L3-SW01		Fortinet FortiSwitch 124F-FPOE	Switch	HQ
VIT-NET-004	HQ-L2-SW01		Cisco Catalyst 2960G 24 Port Switch	Switch	HQ
VIT-NET-005	HQ-L2-SW02		Cisco Catalyst 2960X 48-Port Gigabit PoE Network Switch	Switch	HQ
VIT-WLAN-001	HQ-AP		Linksys EA6300 Advanced Multimedia AC1200 Smart WiFi Wireless Router	Wireless Access Point	HQ
VIT-CMP-001	HQ-PVE01		HP ProBook 445 14 inch G11	Laptop	HQ
VIT-CMP-002	HQ-WS01		HP ProBook 445 14 inch G11	Laptop	HQ
VIT-MET-001	BackupDrive		2 TB Seagate Barracuda in UGREEN USB3 Enclosure	HDD	HQ
VIT-NET-002	BR-Router		Cisco 2951 Integrated Services Router	Router	Branch
VIT-NET-006	BR-L2-SW01		Cisco 2960S Series 48 Port Gigabit Switch	Switch	Branch
VIT-WLAN-002	BR-AP		TP-Link Archer AX23 AX1800 Dual-Band Wi-Fi 6 Router	Wireless Access Point	Branch
VIT-CMP-003	BR-WS01		HP ProBook 445 14 inch G11	Laptop	Branch
VIT CMP-003	DESKTOP-7NFR2KS		HUAWEI MateBook D 15 Laptop	Laptop	Branch

Source inventory from the project asset register

The topology separates each infrastructure role

HEADQUARTERS

BRANCH



HQ performs inter-VLAN routing at the FortiSwitch; the branch uses router-on-a-stick.

VLANs separate users, services, and guests

HQ uses 10.10.0.0/16; the branch uses 10.20.0.0/16. Each VLAN receives a /24 subnet for clear routing and growth.

Site	VLAN / Name	Subnet	Gateway	Purpose
HQ	10 MGMT	10.10.10.0/24	10.10.10.1	Network administration
HQ	20 SERVERS	10.10.20.0/24	10.10.20.1	Centralized VMs and services
HQ	30 / 35	10.10.30.0/24 10.10.35.0/24	.1 on each subnet	Wired users / VantageIT Wi-Fi
HQ	40 / 50	10.10.40.0/24 10.10.50.0/24	.1 on each subnet	IoT-CCTV / guest Wifi
BR	110 MGMT	10.20.10.0/24	10.20.10.1	Branch management
BR	130 / 135	10.20.30.0/24 10.20.35.0/24	.1 on each subnet	Wired users / VantageIT Wi-Fi
BR	150 / 199	10.20.50.0/24 no client subnet	10.20.50.1 / none	Guest Wi-Fi / native-unused

A secure VPN links two routed sites

Only approved traffic crosses the tunnel; each site keeps its own VLAN gateways and routing boundaries.



Branch access is limited to approved HQ services

The VPN provides reachability. ACLs decide what branch users may actually use.

ALLOWED

- AD authentication, DNS, and DHCP relay
- Internal Exchange email
- Department file shares
- Internal web application
- Monitoring and asset tools for IT administrators

BLOCKED OR RESTRICTED

- HQ management VLAN from normal branch clients
- Unapproved access to HQ users VLANs
- Guest Wi-Fi access, internet only
- Network administration outside the management path
- Any traffic not explicitly included in service ACLs

One Proxmox host centralizes six essential services

22 GB

allocated from 32 GB

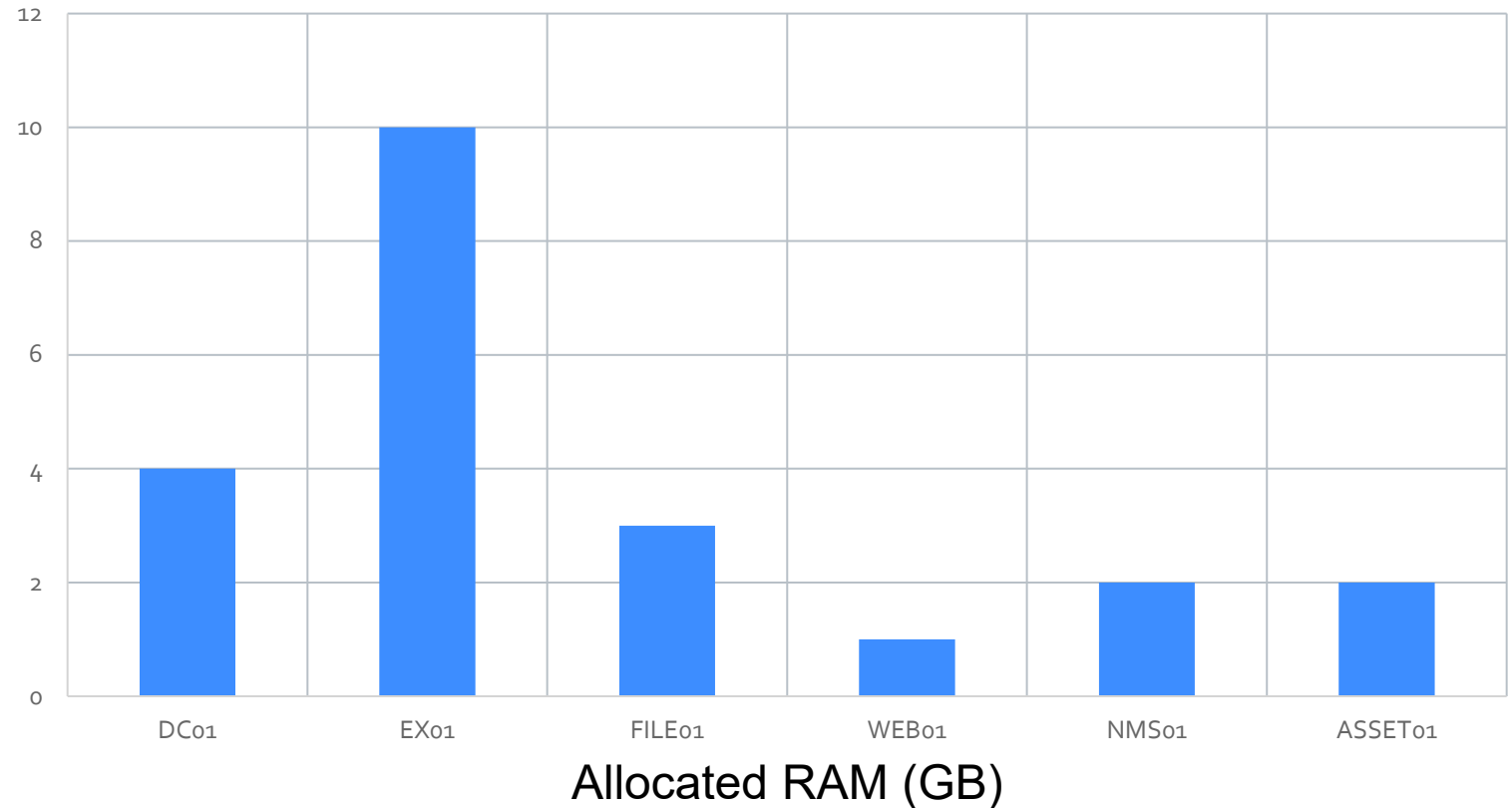
≈10 GB remains for the
Proxmox host and growth.

Windows VM

HQ-DC01 • HQ-EX01

Linux VM

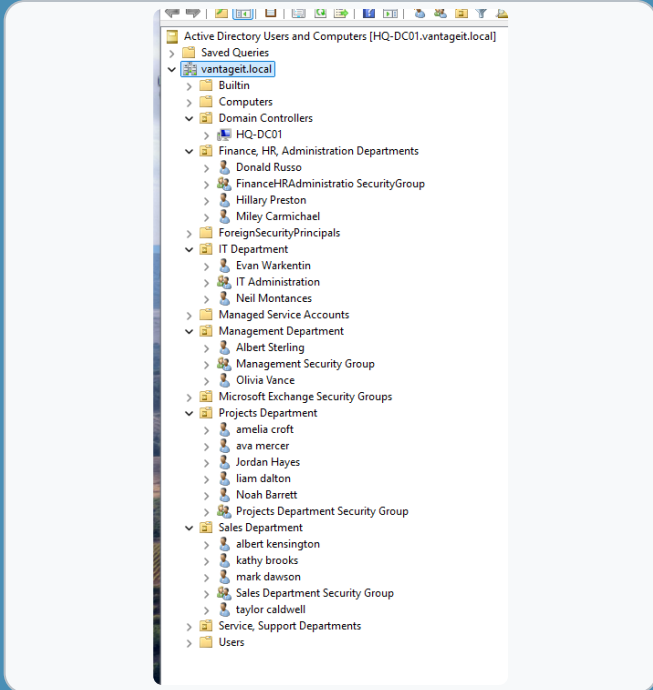
FILE • WEB • NMS • ASSET



Central identity powers internal services

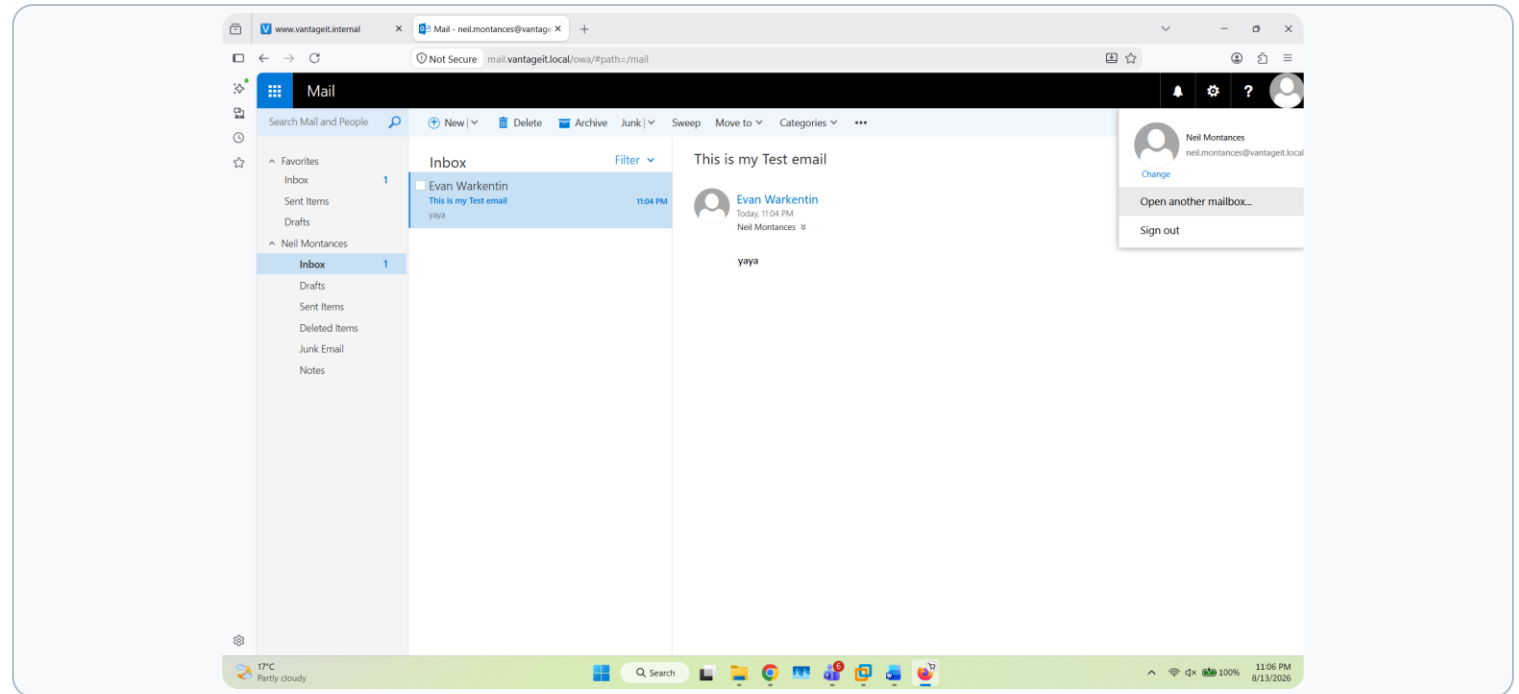
HQ-DC01 provides AD DS, DNS, and DHCP. Exchange and the Linux services use the same centralized network and identity foundation.

ACTIVE DIRECTORY



vantageit.local

INTERNAL EXCHANGE EMAIL

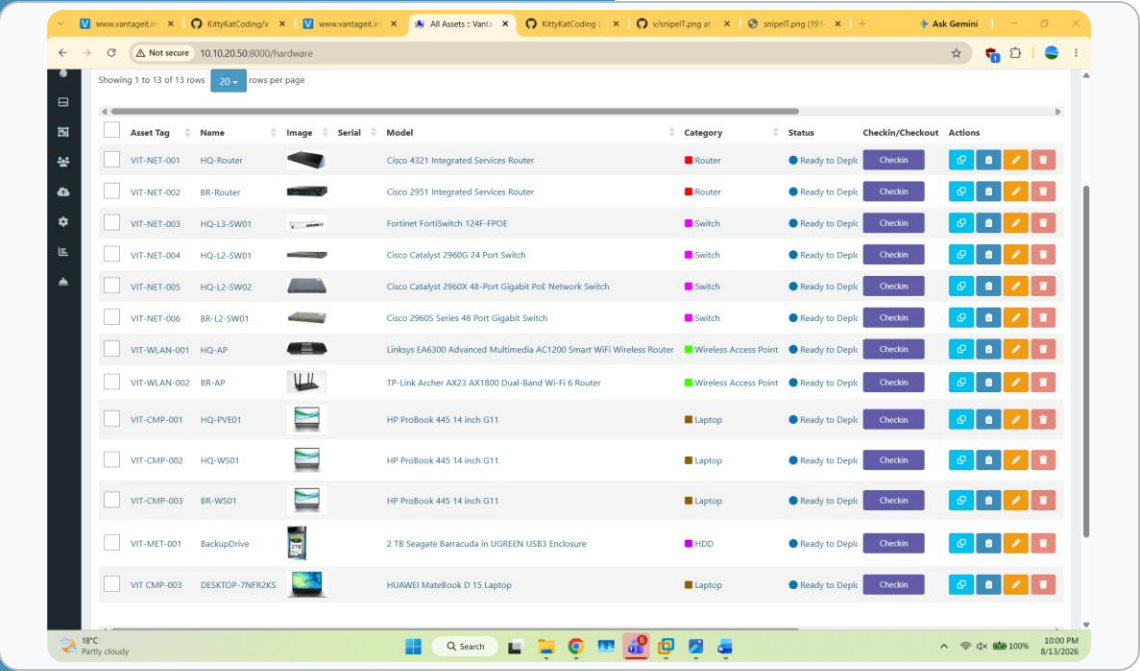


mail.vantageit.local

Inventory and monitoring create visibility

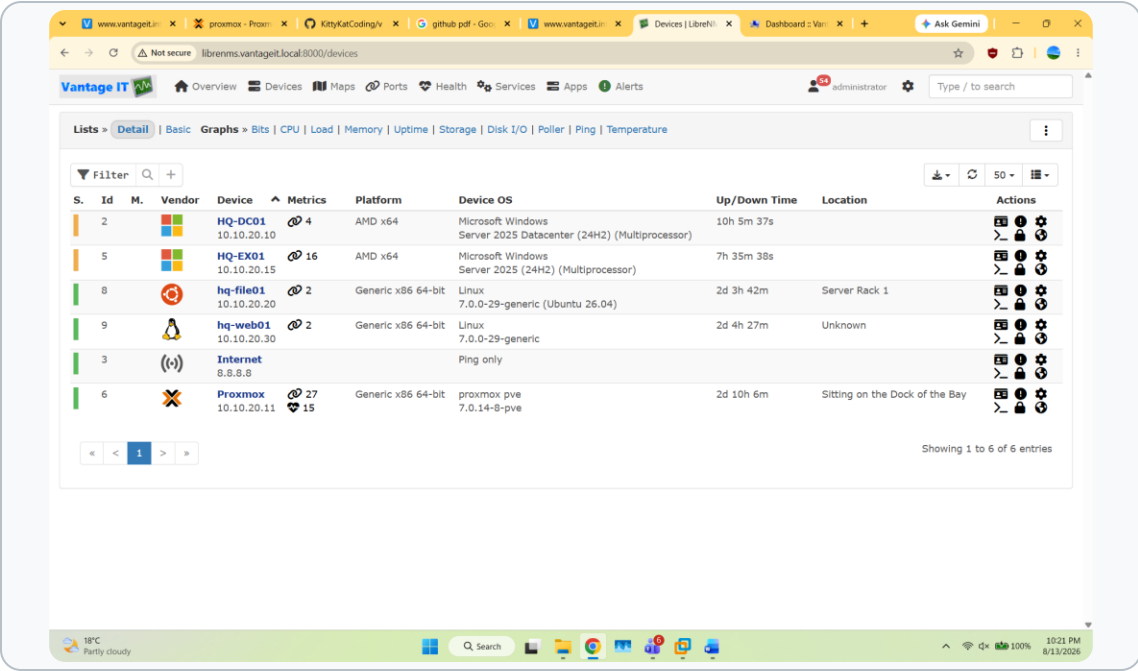
Snipe-IT records what the company owns. LibreNMS shows device health and reachability.

ASSET INVENTORY



Owner • model • category • location • status

NETWORK MONITORING



Availability • device status • SNMP metrics

Security relies on reinforcing layers

The design combines segmentation, least privilege, secure management, and recovery.

IDENTITY

AD groups • separate admin accounts • service-account restrictions

NETWORK

VLAN segmentation • ACLs • management isolation • guest separation

MANAGEMENT

SSH / HTTPS • SNMP • access only from approved paths

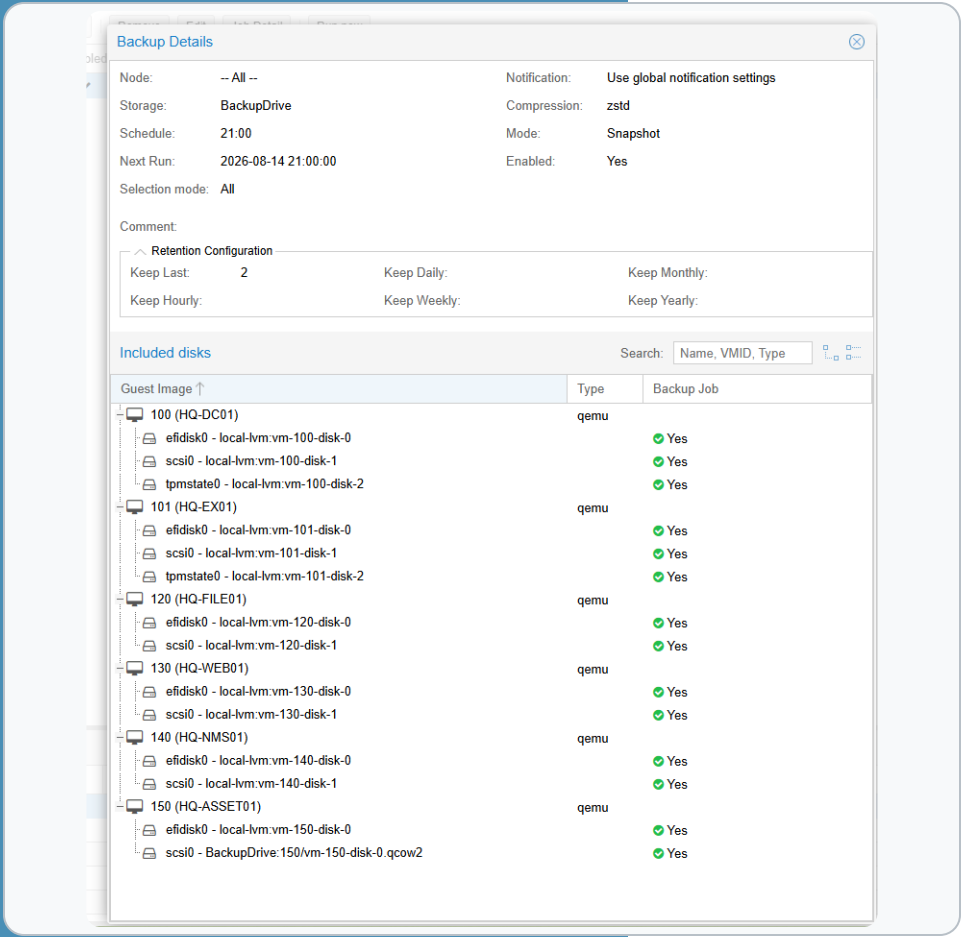
ENDPOINTS

Updates • host firewalls • disabled unused switch ports • BPDU Guard

RECOVERY

Scheduled backups • encrypted media • restore testing • documented ownership

Backups are scheduled, retained, and verified



DAILY

9:00 PM backup job

2 TB

external SATA HDD via USB 3 enclosure

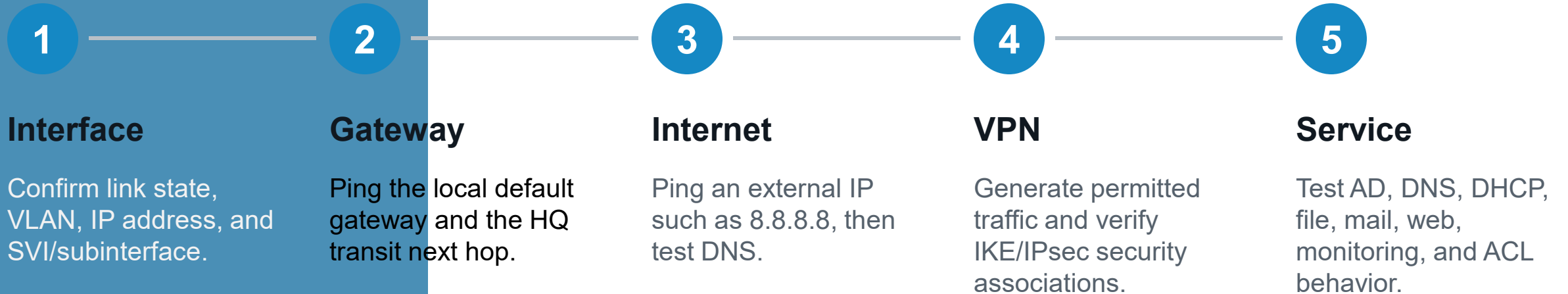
KEEP LAST 2

Retention matches the documented lab storage limit.

Recovery is complete only after a test restore confirms that the backup can be used.

Troubleshooting moves from local to remote

A layered check prevents the team from blaming the VPN or Internet before local routing is proven.



The main risks are practical—and can be reduced

Risk	Risk Description	1-5 likley	1-5 impact	Risk Score (L x I)	Proactive Mitigation Strategy	Reactive Contingency Plan
1	Hardware Failure	2	5	10	Check thermals on all devices	Buy new hardware if it fails
2	Power Interruption	4	5	20	Install an uninterruptable Power supply	Restart all devices and recover from backups if data is lost
3	Storage Failure	1	5	5	check ssd and hdd health data	if the ssd fails, restore proxmox from hdd backup onto new ssd. If backup hdd fails, buy and install new hdd
4	Network Instability	3	3	9	connect everything well with good configs	inspect cables and use pings and tracerouts to see what is wrong
5	Thermal Throttling	2	1	2	Put server laptop onn stand to cool off	place a box fan pointed at the laptop
6	Accidental Disturbance	2	2	4	Make sure no one touches it or I touch it	fix everything that changed

Priority mitigations

- Protect power and cooling.
- Watch disk health and capacity.
- Maintain tested backups and spare configs.

Team contribution

Neil Montances	Switch and router configuration
Evan Warkentin	Virtual machines and Active Directory

A centralized design that stays segmented, secure, and supportable

Segmented

VLANs and least-privilege access

Connected

Secure routing and site-to-site
VPN

Operable

Central services, monitoring,
backup, and testing

The design meets the company's current needs and provides a practical path for controlled growth.